

Presentation4

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1. Basic Idea of ANN An Artificial Neural Network (ANN) is a computational model inspired by the human brain. It consists of interconnected processing units called neurons , which work together to learn patterns from data.

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The ANN algorithm works in two main phases: Forward Propagation Backward Propagation (Training)

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Structure of ANN An ANN consists of: Input Layer Hidden Layer(s) Output Layer Each layer contains neurons connected by weights .

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Mathematical Model of a Neuron Each neuron performs the following computation:

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Variable Meaning (x_i) Input feature (e.g., age, salary, pixel value) (w_i) Weight associated with input (x_i) (n) Number of input features (b) Bias term (shifts activation function) (z) Weighted sum of inputs ($f(z)$) Activation function (y) Output of the neuron

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Forward Propagation (Prediction Phase) Input data $x=(x_1,x_2,..., x_n)$ x is given to the input layer Each neuron calculates weighted sum: $z=\sum w_i x_i +b$ Activation function is applied: $Y=f(z)$ The output of one layer becomes input to the next layer Final layer produces predicted output y

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Common Activation Functions The activation function is also called the transfer function. Activation function is mathematical formulae that determine the output of processing nodes. It transforms inputs into outputs. They introduce a degree of nonlinearity that is valuable for most ANN applications, which governs the relationship between inputs and outputs of a node and a network.

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Activation Function Example : a single neuron with two inputs. Each input is weighted with an appropriate w . The sum of the weighted inputs and the bias forms the input to the transfer function f . Neurons can use any differentiable transfer function f to generate their output.

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Activation Function cont. Each activation function is characterized by its shape, output range, and derivative function. In general, activation functions can be categorized as binary or continuous (differentiable) activation functions.

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Activation Function cont. There are different types of activation functions, the two types are threshold function and sigmoid function. commonly used activation functions are: Tan-Sigmoid, Log-Sigmoid and linear (purelin) activation function, ramp and Gaussian function. See below figure.

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Activation Function cont.

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Activation Function cont. Threshold Function: In engineering, this form of a threshold function is commonly referred to as a Heaviside function

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Activation Function cont. B. Sigmoid Function The sigmoid function, whose graph is "S"-

shaped, is by far the most common form of activation function used in the construction of neural networks. It is defined as a strictly increasing function that exhibits a graceful balance between linear and nonlinear behavior. An example of the sigmoid function is the logistic function defined by

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Loss (Error) Calculation The difference between actual output and predicted output is measured using a loss function. Loss Function Formula Used For Mean Squared Error (MSE) $(y - \hat{y})^2$ Regression Binary Cross-Entropy Binary classification

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Variables Variable Meaning (y) True (actual) output ($\hat{y}$) Predicted output (L) Loss value

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Example Input = (2, 3) = (x_1, x_2) Weights = (0.4, 0.6) = (w_1, w_2) Bias = 0.1
 $z = (2 \times 0.4) + (3 \times 0.6) + 0.1 = 2.7$ Using ReLU: $y = \max(0, 2.7) = 2.7$

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Example Given $W_1=0.4$ $x_1=1$ $W_2=0.6$ $x_2=0$ Actual output=1 Activation Function Sigmoid Calculate Weighted Sum (Net Input)

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Apply Sigmoid Activation

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Compare with Actual Output Actual =1 Predicted output=0.2 Calculate Error (MSE)
Why This Error Matters Large error !' weights need big change Small error !' weights need small change Error is used in backpropagation to update: Weights Bias

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Designing and Training an ANN Design: Architecture of ANN network Determines

number of network weights How many layers and nodes per layer? Structure of artificial neurons Connectivity and choice of activation (transfer) function Training: Ensuring optimum training Learning parameters learning rate, momentum and weight-cost

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Network Design & Training Issues Data preparation The quality of results relates directly to quality of the data(70%). Consolidation, cleaning, selection and preprocessing and Transformation General Steps Get the data set Determine the network topology Training Optimization Testing

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Network Design & Training Issues Learning rate controls how fast ANN learns Momentum smooths learning and speeds convergence Weight cost prevents overfitting by penalizing large weights Proper selection ensures: Fast convergence Stable training Good generalization

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Performance Testing Parameters in ANN After training an ANN, we evaluate it using performance metrics . These metrics depend on whether the task is classification or regression . 1Accuracy_Accuracy measures the percentage of correct predictions . Where: TP – True Positives TN – True Negatives FP – False Positives FN – False Negatives

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Performance Testing Parameters in ANN Precision Precision measures how many predicted positives are actually positive . Important when False positives are costly (e.g., spam detection)

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Performance Testing Parameters in ANN Recall (Sensitivity): measures how many actual positives are correctly identified . Important when Missing a positive is costly (e.g., disease detection)